

● HARD

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Inferences

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Q1

INFORMATION AND IDEAS

Scientists studying Mars long thought the history of its crust was relatively simple. One reason for this is that geologic and climate data collected by a spacecraft showed that the crust was largely composed of basalt, likely as a result of intense volcanic activity that brought about a magma ocean, which then cooled to form the planet's surface. A study led by Valerie Payré focused on additional information—further analysis of data collected by the spacecraft and infrared wavelengths detected from Mars's surface—that revealed the presence of surprisingly high concentrations of silica in certain regions on Mars. Since a planetary surface that formed in a mostly basaltic environment would be unlikely to contain large amounts of silica, Payré concluded that _____blank

Which choice most logically completes the text?

- A. the information about silica concentrations collected by the spacecraft is likely more reliable than the silica information gleaned from infrared wavelengths detected from Mars's surface.
- B. high silica concentrations on Mars likely formed from a different process than that which formed the crusts of other planets.
- C. having a clearer understanding of the composition of Mars's crust and the processes by which it formed will provide more insight into how Earth's crust formed.
- D. Mars's crust likely formed as a result of other major geological events in addition to the cooling of a magma ocean.

The domestic sweet potato (*Ipomoea batatas*) descends from a wild plant native to South America. It also populates the Polynesian Islands, where evidence confirms that Native Hawaiians and other Indigenous peoples were cultivating the plant centuries before seafaring first occurred over the thousands of miles of ocean separating them from South America. To explain how the sweet potato was first introduced in Polynesia, botanist Pablo Muñoz-Rodríguez and colleagues analyzed the DNA of numerous varieties of the plant, concluding that Polynesian varieties diverged from South American ones over 100,000 years ago. Given that Polynesia was peopled only in the last three thousand years, the team concluded that _____ blank

Which choice most logically completes the text?

- A. the cultivation of the sweet potato in Polynesia likely predates its cultivation in South America.
- B. Polynesian peoples likely acquired the sweet potato from South American peoples only within the last three thousand years.
- C. human activity likely played no role in the introduction of the sweet potato in Polynesia.
- D. Polynesian sweet potato varieties likely descend from a single South American variety that was domesticated, not wild.

Ancestral Puebloans, the civilization from which present-day Pueblo tribes descended, emerged as early as 1500 B.C.E. in an area of what is now the southwestern United States and dispersed suddenly in the late 1200s C.E., abandoning established villages with systems for farming crops and turkeys. Recent analysis comparing turkey remains at Mesa Verde, one such village in southern Colorado, to samples from modern turkey populations in the Rio Grande Valley of north central New Mexico determined that the latter birds descended in part from turkeys cultivated at Mesa Verde, with shared genetic markers appearing only after 1280. Thus, researchers concluded that _____ blank

Which choice most logically completes the text?

- A. conditions of the terrains in the Rio Grande Valley and Mesa Verde had greater similarities in the past than they do today.
- B. some Ancestral Puebloans migrated to the Rio Grande Valley in the late 1200s and carried farming practices with them.
- C. Indigenous peoples living in the Rio Grande Valley primarily planted crops and did not cultivate turkeys before 1280.
- D. the Ancestral Puebloans of Mesa Verde likely adopted the farming practices of Indigenous peoples living in other regions.

Mosses can struggle in harsh desert conditions because these plants require enough sunlight for photosynthesis but not so much that they risk drying out. Researchers Jenna Ekwealor and Kirsten M. Fisher found several species of *Syntrichia caninervis*, a type of desert moss, growing under quartz crystals in California's Mojave Desert. To evaluate whether these semitransparent rocks benefited the moss, the researchers compared the shoot tissue, a measure of plant growth, of *S. caninervis* when growing on the soil surface versus when the moss was growing under the quartz rocks. They found that the shoot tissue was 62% longer for moss growing under the quartz as compared to moss on the soil surface, suggesting that _____ blank

Which choice most logically completes the text?

- A. *S. caninervis* is one of the few types of moss that can survive under semitransparent rocks.
- B. quartz crystals do not transmit the necessary sunlight for photosynthesis in *S. caninervis*.
- C. *S. caninervis* growing under quartz crystals experience lower light intensity and are thus able to retain more moisture.
- D. quartz crystals are capable of supporting *S. caninervis* growth if the crystals are not too thin.

Researchers recently found that disruptions to an enjoyable experience, like a short series of advertisements during a television show, often increase viewers' reported enjoyment. Suspecting that disruptions to an unpleasant experience would have the opposite effect, the researchers had participants listen to construction noise for 30 minutes and anticipated that those whose listening experience was frequently interrupted with short breaks of silence would thus _____ blank

Which choice most logically completes the text?

- A. find the disruptions more irritating as time went on.
- B. rate the listening experience as more negative than those whose listening experience was uninterrupted.
- C. rate the experience of listening to construction noise as lasting for less time than it actually lasted.
- D. perceive the volume of the construction noise as growing softer over time.

Geoglyphs are large-scale designs of lines or shapes created in a natural landscape. The Nazca Lines were created in the Nazca Desert in Peru by several Indigenous civilizations over a period of many centuries. Peruvian archaeologist Johnny Isla specializes in these geoglyphs. At a German exhibit about the Nazca Lines, he saw an old photograph of a large geoglyph of a whalelike figure and was surprised that he didn't recognize it. Isla returned to Peru and used a drone to search a wide area, looking for the figure from the air. This approach suggests that Isla thought that if he hadn't already seen it, the whalelike geoglyph _____ blank

Which choice most logically completes the text?

- A. must represent a species of whale that went extinct before there were any people in Peru.
- B. is actually located in Germany, not Peru, and isn't part of the Nazca Lines at all.
- C. is probably in a location Isla hadn't ever come across while on the ground.
- D. was almost certainly created a long time after the other Nazca Lines geoglyphs were created.

Compiled in the late 1500s largely through the efforts of Indigenous scribes, *Cantares Mexicanos* is the most important collection of poetry in Classical Nahuatl, the principal language of the Aztec Empire. The poems portray Aztec society before the occupation of the empire by the army of Spain, and marginal notes in *Cantares Mexicanos* indicate that much of the collection's content predates the initial invasion. Nonetheless, some of the poems contain inarguable references to beliefs and customs common in Spain during this era. Thus, some scholars have concluded that _____ blank

Which choice most logically completes the text?

- A. while its content largely predates the invasion, *Cantares Mexicanos* also contains additions made after the invasion.
- B. although those who compiled *Cantares Mexicanos* were fluent in Nahuatl, they had limited knowledge of the Spanish language.
- C. before the invasion by Spain, the poets of the Aztec Empire borrowed from the literary traditions of other societies.
- D. the references to beliefs and customs in Spain should be attributed to a coincidental resemblance between the societies of Spain and the Aztec Empire.

Tides can deposit large quantities of dead vegetation within a salt marsh, smothering healthy plants and leaving a salt panne—a depression devoid of plants that tends to trap standing water—in the marsh’s interior. Ecologist Kathryn Beheshti and colleagues found that burrowing crabs living within these pannes improve drainage by loosening the soil, leading the pannes to shrink as marsh plants move back in. At salt marsh edges, however, crab-induced soil loosening can promote marsh loss by accelerating erosion, suggesting that the burrowing action of crabs _____ blank

Which choice most logically completes the text?

- A. can be beneficial to marshes with small pannes but can be harmful to marshes with large pannes.
- B. may promote increases in marsh plants or decreases in marsh plants, depending on the crabs’ location.
- C. tends to be more heavily concentrated in areas of marsh interiors with standing water than at marsh edges.
- D. varies in intensity depending on the size of the panne relative to the size of the surrounding marsh.

The ratio of methane to other atmospheric constituents—represented by a measure called the methane mole fraction—influences a variety of meteorological phenomena, notably precipitation and humidity. For Titan, Saturn’s largest moon, the observational data that exist are too sparse and discrepant to fully constrain the range of the methane mole fraction at various atmospheric levels. Juan Lora and colleagues point out that outputs of the IPSL atmospheric model of Titan, which track closely to observations in some respects, reflect how the model’s developers responded to this challenge: by prescribing a uniform methane mole fraction for the lowest level of the atmosphere. It is therefore important to note that _____ blank

Which choice most logically completes the text?

- A. some disagreements between the model’s simulations of Titan’s precipitation and humidity and the moon’s actual precipitation and humidity are to be expected.
- B. further observations of Titan may clarify the moon’s methane mole fraction sufficiently for the model to employ a single value rather than a range.
- C. even though the model’s outputs sometimes agree with observational data, Titan’s real methane mole fraction is likely higher than the methane mole fraction used in the model.
- D. inconsistencies across the model’s simulations of Titan’s precipitation and humidity could be attributable to variations in the moon’s methane mole fraction.

One theory behind human bipedalism speculates that it originated in a mostly ground-based ancestor that practiced four-legged “knuckle-walking,” like chimpanzees and gorillas do today, and eventually evolved into moving upright on two legs. But recently, researchers observed orangutans, another relative of humans, standing on two legs on tree branches and using their arms for balance while they reached for fruits. These observations may suggest that _____ blank

Which choice most logically completes the text?

- A. bipedalism evolved because it was advantageous to a tree-dwelling ancestor of humans.
- B. bipedalism must have evolved simultaneously with knuckle-walking and tree-climbing.
- C. moving between the ground and the trees would have been difficult without bipedalism.
- D. a knuckle-walking human ancestor could have easily moved bipedally in trees.